parts in a million. However, if the Sun kept its same mass and were made smaller and smaller in circumference, then the curvature inside it would become stronger and stronger, the downward dip of the bowl in the embedding diagram of Figure 3.3 would become more and more pronounced, and the ratio of circumference to diameter would become substantially less than π.

Because space is different in different reference frames ("your space is a mixture of my space and my time, if we move relative to each other"), the details of the star's spatial curvature will be different as measured in a reference frame that moves at high speed relative to the star than as measured in a frame where the star is at rest. In the space of the high-speed reference frame, the star is somewhat squashed perpendicular to its direction of motion, so the embedding diagram looks much like that of Figure 3.3, but with the bowl compressed transversely into an oblong shape. This squashing is the curved-space variant of the contraction of space that Fitzgerald discovered in a universe without gravity (Chapter 1).

Schwarzschild's solution to the Einstein field equation describes not only this curvature (or warpage) of space, but also a warpage of time near the star—a warpage produced by the star's strong gravity. In a reference frame that is at rest with respect to the star, and not flying past it at high speed, this time warpage is precisely the *gravitational time dilation* discussed in Chapter 2 (Box 2.4 and associated discussion): Near the star's surface, time flows more slowly than far away, and at the star's center, it flows slower still.

In the case of the Sun, the time warpage is small: At the Sun's surface, time should flow more slowly by just 2 parts in a million (64 seconds in one year) than far from the Sun, and at the Sun's center it should flow more slowly than far away by about 1 part in 100,000 (5 minutes in one year). However, if the Sun kept its same mass and were made smaller in circumference so its surface was closer to its center, then its gravity would be stronger, and correspondingly its gravitational time dilation—its warpage of time—would become larger.

One consequence of this time warpage is the *gravitational redshift* of light emitted from a star's surface. Since the light's frequency of oscillation is governed by the flow of time at the place where the light is emitted, light emerging from atoms on the star's surface will have a lower frequency when it reaches Earth than light emitted by the same kinds of atoms in